"The Plane"

F **Definition:** A plane is a surface such that if any two points are taken on it, the st. line joining them, lies wholly on the surface.

F General eqn of plane:

$$ax + by + cz + d = 0$$

& Intersect form: $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1.$

F Eqn of a plane passes through (x_1, y_1, z_1) is

$$a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$$

F Eqn of the plane passes through (x_1, y_1, z_1), (x_2, y_2, z_2) & (x_3, y_3, z_3) is

$$\begin{vmatrix} x - x_1 & y - y_1 & z - z_1 \\ x_1 - x_2 & y_1 - y_2 & z_1 - z_2 \\ x_2 - x_3 & y_2 - y_3 & z_2 - z_3 \end{vmatrix} = 0$$

F Angle between two plane

$$a_1x + b_1y + c_1z + d_1 = 0$$

$$a_2x + b_2y + c_2z + d_2 = 0$$

$$\cos\theta = \pm \frac{a_1a_2 + b_1b_2 + c_1c_2}{\sqrt{a_1^2 + b_1^2 + c_1^2} \cdot \sqrt{a_2^2 + b_2^2 + c_2^2}}$$

If perpendicular then, $a_1a_2 + b_1b_2 + c_1c_2 = 0$

& if parallel, then, $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$.

☐ Passes through the point and parallel to the plane is $ax+by+cz+k=0$

☐ Passes through two points (x_1, y_1, z_1); (x_2, y_2, z_2) and perpendicular with $ax+by+cz+d=0$, then.

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ x_1-x_2 & y_1-y_2 & z_1-z_2 \\ a & b & c \end{vmatrix} = 0$$

☐ Passes through a point ~~and per~~ (x_1, y_1, z_1) and perpendic with $a_1x+b_1y+c_1z+d_1=0$ and $a_2x+b_2y+c_2z+d_2=0$, th

$$\begin{vmatrix} x-x_1 & y-y_1 & z-z_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix} = 0$$

1. Find the eq^n of the plane passes through the point $(2,3,1)$, $(1,1,3)$ & $(2,2,3)$. Find also the perpendicular distance from the point $(5,6,7)$ to this plane.

Sol^n: Eq^n of the plane passes through the given point is;

$$\begin{vmatrix} x & y & z & 1 \\ 2 & 3 & 1 & 1 \\ 1 & 1 & 3 & 1 \\ 2 & 2 & 3 & 1 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} x-2 & y-3 & z-1 \\ 1 & 2 & -2 \\ -1 & -1 & 0 \end{vmatrix} = 0$$

$$\Rightarrow (x-2)(0-2) - (y-3)\{\cancel{(1+2)}-2-0\} + (z-1)(-1+2) = 0$$

$$\Rightarrow -2x+4+2y+6+z-1=0$$

$$\Rightarrow 2x-2y-z+3=0$$

2nd part: Perpendicular distance from the point $(5,6,7)$ to the plane is

$$\left|\frac{2\times5-2\times6-7+3}{\sqrt{4+4+1}}\right|$$

$$= \left|\frac{-6}{\pm3}\right| = 2$$

Formula

$$p.d = \frac{aa_1+bb_1+cc_1}{\sqrt{a^2+b^2+c^2}}$$

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2. Find the equ^n of the plane through the points $(2,2,1)$ and $(9,3,6)$ and perpendicular to the plane $2x+6y+6z=9$.

Sol^n: Equ^n of the plane through the point $(2,2,1)$ is

$$a(x-2)+b(y-2)+c(z-1)=0 \quad \text{---(I)}$$

If the plane passes through the $(9,3,6)$ is

$$a(9-2)+b(3-2)+c(6-1)=0$$

$$\Rightarrow 7a+b+5c=0 \quad \text{---(II)}$$

As the plane perpendicular with the given plane so,

$$2a+6b+6c=0 \quad \text{---(IV)}$$

From (II) & (III), we get,

$$\frac{a}{6-30}=\frac{b}{10-42}=\frac{c}{42-2}$$

$$\Rightarrow \frac{a}{-24}=\frac{b}{-32}=\frac{c}{40}$$

$$\Rightarrow \frac{a}{3}=\frac{b}{4}=\frac{c}{-5}=k \text{ (say)}$$

$\therefore a=3k,\ b=4k,\ c=-5k.$

Putting this values of a,b,c in (I) we get,

$$3k(x-2)+4k(y-2)-5k(z-1)=0$$

$$\Rightarrow 3x+4y-5z-9=0.$$

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3\. Find the eqn of the plane which perpendicular with the plane $5x+3y+6z+8=0$ and containe the line through the intersection of the planes $x+2y+3z+4=0$ and $2x+y-z+5=0$.

Soln:

Eqn of the plane containing the line through the intersection of the plane $x+2y+3z-4=0$ and $2x+y-z+5=0$ is.

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$$(x+2y+3z-4)+k(2x+y-z+5)=0 \quad \text{---} (1)$$

$$\Rightarrow (1+2k)x+(2+k)y+(3-k)z+(5k-4)=0$$

if it is perpendicular with $5x+3y+6z+8=0$

then,

$$5(1+2k)+3(2+k)+6(3-k)=0$$

$$\Rightarrow 5+10k+6+3k+18-6k=0$$

$$\Rightarrow 7k+29=0$$

$$\therefore k=-29/7$$

Putting this value of k in (1) we get,

$$x+2y+3z-4-29/7(2x+y-z+5)=0$$

$$\Rightarrow 7x+14y+21z-28-58x-29y+29z-145=0$$

$$\Rightarrow -51x-15y+50z-173=0$$

$$\therefore 51x+15y-50z+173=0$$ which is the required equn of plane.